Perceptron Trick

How to label regions?

Line eq^n = $Ax + By + C = 0$

If value of

$C \uparrow \Rightarrow$ line goes down (Paralelly)

$C \downarrow \Rightarrow$ line goes up (Paralelly)

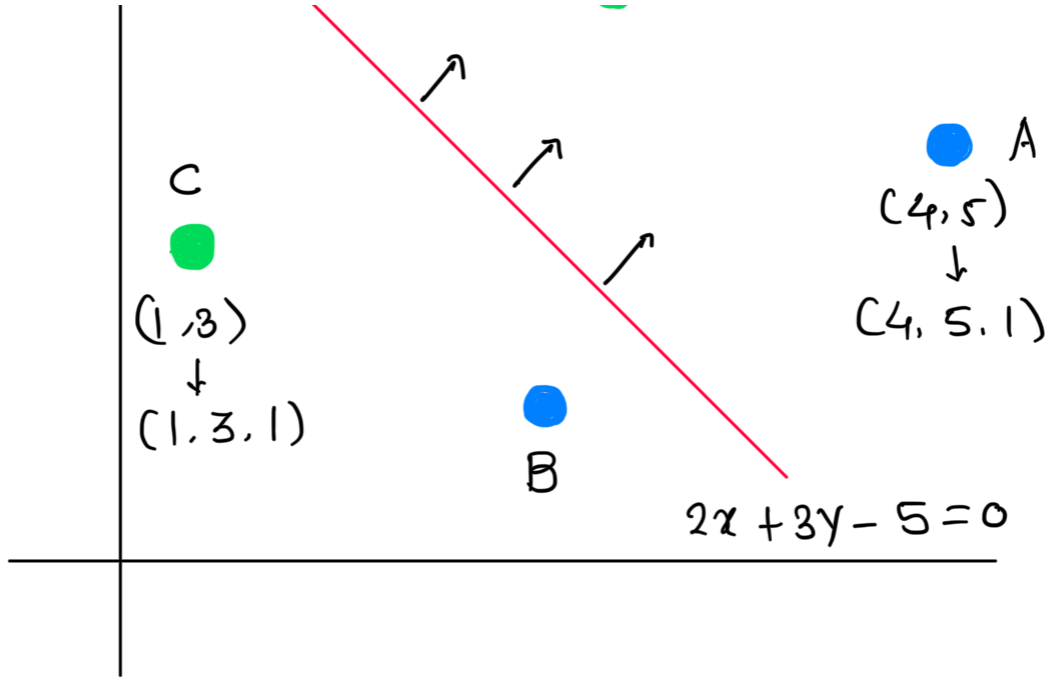
$A \uparrow \Rightarrow$ Rotates upward (clockwise)

$A \downarrow \Rightarrow$ Rotates downward (Anticlockwise)

$B \uparrow \Rightarrow$ Rotates downward (Anticlockwise)

$B \downarrow \Rightarrow$ Rotates upward (clockwise)

Transformations



For point A we want line to move to point A i.e. in +ve region

$$\begin{array}{rccc} & 2 & 3 & 5 \\ - & 4 & 5 & 1 \\ \hline \end{array}$$

$$-2x - 2y + 4 = 0$$

For point C, we are moving line in -ve region

$$\begin{array}{rccc} & 2 & 3 & 5 \\ + & 1 & 3 & 1 \\ \hline \end{array}$$

$$3x + 6x + 6 = 0$$

Learning Rate is applied before adding or subtracting points from line.

Simplified Algo

$$x_i \in N \text{ and } \sum w_i x_i \geq 0$$

$$w_n = w_0 - \eta x_i$$

$$x_i \in P \text{ and } \sum w_i x_i < 0$$

$$w_n = w_0 + \eta x_i$$